

$$\begin{aligned}
 \textcircled{2} \quad & \int_0^{\pi/2} \sqrt{\tan x} \, dx = ? \\
 & = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\cos x}} \, dx = \int_0^{\pi/2} (\sin x)^{1/2} (\cos x)^{-1/2} \, dx \\
 & \quad 2m-1=1/2 \quad m=3/4 \\
 & \quad 2n-1=-1/2 \quad n=1/4 \\
 & = \frac{1}{2} B\left(\frac{3}{4}, \frac{1}{4}\right) = \frac{\pi}{\sin \frac{3\pi}{4}} = \pi \cdot \frac{1}{\sqrt{2}}
 \end{aligned}$$

$$\textcircled{3} \quad \int_0^1 x^m (\ln x)^n \, dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$$

$$\ln x = -u \Rightarrow x = e^{-u}$$

$$\int_0^1 x^m (\ln x)^n \, dx = \int_{\infty}^0 e^{-u(m+1)} (-u)^n (-du) = \int_0^{\infty} e^{-u(m+1)} (-u)^n \, du$$

$$= (-1)^n \int_0^{\infty} e^{-u(m+1)} u^n \, du$$

$$\begin{aligned}
 u(m+1) &= t \\
 du &= \frac{dt}{m+1}
 \end{aligned}$$

$$= (-1)^n \int_0^{\infty} e^{-t} \cdot \frac{t^n}{(m+1)^n} \cdot \frac{dt}{m+1}$$

$$= \frac{(-1)^n}{(m+1)^{n+1}} \underbrace{\int_0^{\infty} t^n e^{-t} \, dt}_{n!}$$

Sonuç sağlandı!